

Session 2 · Deck 2.B1

Krusell–Smith: The Distribution as a State

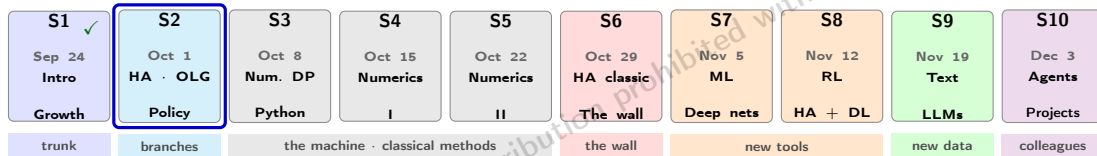
Zhigang Feng

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Where We Are

▼ you are here: 2.B1



Build it on paper → solve it on the machine → push it to the frontier with AI

Block B: 2.B1 aggregate risk and the distribution as a state · 2.B2 finite lives and cohorts · 2.B3 the government as a player: Ramsey and time consistency.

A Change of Status for One Object

Block A. λ^* is an *unknown we solve for, once*. Prices are numbers. The household forecasts nothing.



add a TFP shock

Block B. Γ_t is a *state variable that moves every period*. Prices move with it. The household must forecast **a distribution**.

Count the dimensions

Discretize wealth on a modest grid of 100 points and employment on 2. The distribution is then a point in a **200-dimensional simplex**.

In Block A we solved for *one* such point. Here we need a *function* of it — which is the next slide.

Krusell & Smith (1998, *JPE*) found that, in the standard calibration, **one number** does almost all the work. Understanding why — and where it stops being true — is this deck.

The Environment: Two Kinds of Shock

- **Aggregate productivity** $Z_t \in \{Z_g, Z_b\}$, a two-state Markov chain. Output is

$$Y_t = Z_t F(K_t, \bar{L}_t).$$

- **Idiosyncratic employment** $\varepsilon_t \in \{0, 1\}$. The pair (ε_t, Z_t) follows a *joint* Markov chain: unemployment is higher in the bad aggregate state, $u_b > u_g$.
- Households hold capital a with $a' \geq 0$, and supply $\bar{\ell}$ units of labour when employed.
- Factor prices:

$$r_t = Z_t F_K(K_t, \bar{L}_t) - \delta, \quad w_t = Z_t F_L(K_t, \bar{L}_t).$$

The correlation matters

Idiosyncratic and aggregate risk are *not* independent: you are more likely to lose your job exactly when everyone else is losing theirs, and when assets pay less.

That correlation is why a law of large numbers no longer removes aggregate uncertainty. Individual risk still averages out *within* a period — but the period itself is random.

One Distribution, or a Distribution of Distributions?

Aiyagari: finite-dimensional

Fix the bins — N asset points, n_z shock states. Then λ is a point in the simplex $\Delta^{Nn_z-1} \subset \mathbb{R}^{Nn_z}$, and the whole model asks for **one such point**, together with one number r^* .

- On the 100×2 grid: 199 free numbers plus a price — a root-find in $\mathbb{R}^{200}$.
- That the point exists and is unique is 2.A3. That is all “steady state” means here.

Aiyagari asks for a point in the simplex. Krusell–Smith asks for a function on it. The 199 numbers were never the problem — needing a function *of* them is.

Krusell–Smith: infinite-dimensional

Γ_t moves. The unknown is no longer a point in that simplex but a **function on it** — the value $V(a, \varepsilon, Z, \Gamma)$ and the law of motion H .

- Equivalently, the object with a stationary description is a **measure over measures** μ^* : stationary when $\int f d\mu^* = \int Tf d\mu^*$ for every test function f .
- The same fixed-point idea, one level up.

Why the Household Cannot Avoid Forecasting Γ

To choose a' today, the household needs the return on saving *tomorrow*, r_{t+1} . But

$$r_{t+1} = Z_{t+1} F_K(K_{t+1}, \bar{L}_{t+1}) - \delta,$$

$$K_{t+1} = \int a \, d\Gamma_{t+1}.$$

So forecasting the price requires forecasting K_{t+1} , which requires forecasting Γ_{t+1} , which depends on **everyone else's saving decision** — which depends on *their* forecast.

This is the consistency requirement of the RCE template from 1.B1, now at its most demanding: the object being forecast is an infinite-dimensional measure.

Contrast with 1.B1

In the representative-agent growth model the household forecast $K' = H(K, z)$ — one number to one number.

Here the analogous object is $\Gamma' = H(\Gamma, Z, Z')$: a map from measures to measures.

Note H depends on Z' as well as Z : next period's distribution depends on which aggregate state actually realizes.

Recursive Competitive Equilibrium with Aggregate Risk

The individual state is (a, ε) ; the aggregate state is (Z, Γ) .

The household solves

$$V(a, \varepsilon, Z, \Gamma) = \max_{c, a'} \left\{ u(c) + \beta \mathbb{E}[V(a', \varepsilon', Z', \Gamma') \mid \varepsilon, Z] \right\}$$

subject to

$$c + a' = w(Z, K) \bar{\ell} \varepsilon + (1 + r(Z, K)) a,$$

$$a' \geq 0, \quad K = \int a d\Gamma,$$

$$\Gamma' = H(\Gamma, Z, Z')$$

Equilibrium

V and $a'(\cdot)$ are optimal given H ; prices are marginal products; **and H is generated by $a'(\cdot)$ itself** — the distribution that households' own choices produce is the one they forecast.

A fixed point in a space of **operators on measures**. Writing it down is easy; there is no hope of solving it directly.

The Krusell–Smith Bet

Step 1: replace Γ by a few moments.

$$m_j = \int a^j d\Gamma, \quad j = 1, \dots, J,$$

and let households condition on $m = (m_1, \dots, m_J)$ instead of on Γ . In practice $J = 1$, so $m_1 = K$.

Step 2: postulate a law of motion for those moments. The standard choice is log-linear and state-dependent:

$$\log K' = A(Z) + B(Z) \log K.$$

Four unknown numbers:

$$A(Z_g), B(Z_g), A(Z_b), B(Z_b).$$

What just happened

The state fell from $(a, \varepsilon, Z, \Gamma)$ — infinite-dimensional — to (a, ε, Z, K) , which is **four** dimensions, two of them discrete.

That is an ordinary dynamic program.

It is a **bet**, not a theorem. Households are assumed to use a forecasting rule that is not, in general, the true one. This is bounded rationality with a consistency check.

Closing the Loop, and the Number That Made It Famous

The rule must be the one households' own behaviour generates:

1. Guess $\{A(Z), B(Z)\}$.
2. Solve the household problem given that rule.
3. Simulate a large panel for many periods; record the realized $\{K_t, Z_t\}$.
4. Regress $\log K_{t+1}$ on $\log K_t$, state by state.
5. If the coefficients differ from the guess, update and repeat.

We do not run this today. The algorithm, its accuracy tests, and what breaks are S6.A2. Here we only need to know it *closes*.

Approximate aggregation

At the fixed point, the forecasting regression has

$$R^2 > 0.9999.$$

Mean capital alone predicts next period's mean capital almost perfectly. The other 199 dimensions of Γ are nearly irrelevant *for prices*.

This is the result that made heterogeneous-agent business-cycle models feasible for twenty years.

Why Does It Work? (And Why That Is Not Reassuring)

The mechanism. Aggregate capital is $\int a d\Gamma$. It is insensitive to the shape of Γ provided the saving policy $a'(a, \varepsilon)$ is **nearly linear in** a over the region where most wealth sits.

- Rich households hold most of the capital, and they are far from the borrowing constraint.
- Away from the constraint, the policy function is close to linear.
- So redistributing wealth among the rich barely moves aggregate saving — and prices only see the aggregate.

The caveat that matters

Approximate aggregation is a property of the **calibration**, not of the model class.

It weakens wherever the constraint binds for households who hold real wealth: fatter tails, larger idiosyncratic risk, tighter limits, illiquid assets (HANK).

And note what it says: the distribution does not matter *for forecasting prices*. It says nothing about whether the distribution matters for **welfare** or **policy**.

Where the Moment Approach Runs Out

- **No guidance on functional form.** Why log-linear? Why these moments? Adding m_2, m_3 helps a little and costs a dimension each.
- **R^2 is the wrong test.** A near-perfect forecasting regression can coexist with real errors in individual decisions. Den Haan (2010) gives accuracy tests that are harder to pass.
- **Welfare and inequality need the shape,** not the mean. Moments chosen to forecast prices are not moments chosen to measure who gains from a policy.
- **Optimal policy needs more still.** A government choosing taxes needs to know how the **whole distribution** responds — effectively derivatives of V with respect to Γ . No finite set of moments delivers that.

The thread to pull

The last point is not a technicality. It is precisely why optimal taxation with heterogeneous agents stayed open long after Krusell–Smith made business cycles tractable.

Where this goes: S6.A2 solves Krusell–Smith properly and measures the errors; S6.A3 shows why optimal policy hits a wall. First, though, a second literature has been working on exactly this object.

Krusell–Smith Is a Mean-Field Game

This deck	Mean-field games
continuum of households	continuum of players
household Bellman, backward	HJB equation
law of motion of Γ , forward	Kolmogorov forward eq. (KFE)
employment shock ε	idiosyncratic noise
aggregate shock Z	common noise
H generated by $a'(\cdot)$ itself	equilibrium: the fixed point
$V(a, \varepsilon, Z, \Gamma)$	value in the master equation

The name is new; the object is not. **The recursive competitive equilibrium with aggregate risk is already a master equation**, written in discrete time.

Two cases of one object

Aiyagari is a stationary MFG *without* common noise: the distribution settles at one point λ .

Krusell–Smith is an MFG *with* common noise: Z hits everyone at once, so Γ_t stays random and must enter the state.

Two literatures, one fixed point

Named in engineering and mathematics: Huang, Malhamé & Caines (2006); Lasry & Lions (2007). Macro had the object first (Aiyagari 1994; Krusell & Smith 1998); Achdou, Han, Lasry, Lions & Moll (*REStud* 2022) brought the language back.

Common Noise Is Why Γ Must Be a State

Without common noise the path $\{\Gamma_t\}$ is deterministic. HJB backward and KFE forward *in time* suffice — how transitions and “MIT” shocks are solved.

With common noise the path is random. In continuous time, with Z switching at rate η_Z and $\partial_t \Gamma = \mathcal{K}_Z \Gamma$ the KFE, the value satisfies

$$\begin{aligned} \rho V = & \max_c \{u(c) + \partial_a V s\} + [\varepsilon \text{ jumps}] \\ & + \eta_Z [V(\cdot, Z', \Gamma) - V(\cdot, Z, \Gamma)] \\ & + \int \frac{\delta V}{\delta \Gamma}(x) (\mathcal{K}_Z \Gamma)(x) dx. \end{aligned}$$

The last term is the **capital gain from the distribution moving**: a PDE on the space of measures.

The Krusell–Smith bet, restated

Let V depend on Γ only through $K = \int a d\Gamma$. That replaces the derivative $\delta V / \delta \Gamma$ — a whole function — by one number: a **projection of the master equation** onto one moment.

What the theory adds

- Uniqueness of equilibrium under the Lasry–Lions **monotonicity** condition.
- With common noise, the master equation is well posed and finite-player equilibria converge to the MFG — which licenses step 3’s **finite** panel (Carmona & Delarue 2018; Cardaliaguet, Delarue, Lasry & Lions 2019).

What Recent Mean-Field Work Offers for Solving It

1. Local: linearize the master equation

Expand around the stationary distribution: all of Γ , but small shocks only. At first order the master equation becomes a Bellman equation for the derivatives of V in Γ (Bilal 2023). Kin: Reiter (2009); Ahn, Kaplan, Moll, Winberry & Wolf (2018).

3. The outer loop is a learning dynamic

Step 5's damped update is a form of **fictitious play**. MFG theory proves when such loops converge: potential games (Cardaliaguet & Hadikhanloo 2017); monotone games, by online mirror descent (Pérolat et al. 2022).

2. Global: finite population, learned V

Replace Γ by N agents or a finite basis, and learn V with a network that is **invariant to relabelling agents** (Germain, Laurière, Pham & Warin 2022). Gu, Laurière, Merkel & Payne (2024) solve continuous-time HA master equations this way. Fernández-Villaverde, Hurtado & Nuño (ECMA 2023) replace the log-linear rule with a neural network and find the law of motion nonlinear.

Each answers a bullet of “Where the Moment Approach Runs Out”: functional form, the shape of Γ , and a test stricter than R^2 — the master-equation residual checks all of V .

Takeaways

- With aggregate risk, the wealth distribution stops being an unknown to solve for and becomes a **state variable** — one the household must forecast to price its own saving.
- The recursive equilibrium is a fixed point in a space of maps from measures to measures: well defined, and directly unsolvable.
- **Krusell–Smith** replaces Γ by its mean and postulates $\log K' = A(Z) + B(Z) \log K$, then requires the rule to reproduce itself in simulation.
- **Approximate aggregation** ($R^2 > 0.9999$) makes this work — because saving policies are nearly linear where the wealth is. It is a feature of the calibration, and it degrades exactly where constraints bind.
- Forecasting prices well is **not** the same as representing the distribution well. Welfare, inequality and optimal policy all need more.
- Krusell–Smith is a **mean-field game with common noise**; its moment rule is one projection of the master equation. That literature offers local (linearized) and global (finite-population, symmetric-network) alternatives.

Bridge: A Different Reason the Distribution Moves

In Krusell–Smith the distribution moves because an **aggregate shock** hits it. Everyone is infinitely lived; the cross-section churns but nobody enters or leaves.

Now remove that assumption. If households are **born and die**, then:

- age becomes part of the individual state, and the distribution splits into **cohorts**;
- wealth follows a hump over the life cycle, so the cross-section is a mixture of ages, not a stationary pool;
- and — the surprise — a competitive equilibrium can be **Pareto inefficient** even with no shocks, no frictions, and complete markets within each period.

Next (2.B2): finite lives

The quantitative life-cycle household — survival risk, retirement, age-varying productivity — and the equilibrium with cohort distributions.

Plus the two-period example that shows why saving too much can be a genuine failure, and what social security has to do with it.